Project Proposal: Developing Vibrotactile Haptics Virtual Reality Gloves to alleviate user experience

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1 Introduction

Virtual Reality (VR) has revolutionized digital interaction, but one significant limitation remains—the lack of tactile feedback. This project aims to develop vibrotactile haptic gloves that integrate seamlessly with VR, allowing users to physically feel virtual objects. By embedding mini vibration motors at the fingertips of the gloves and connecting them to a VR game via an ESP32 module, I aim to enable real-time haptic feedback. When a user touches an object in the VR world, the game sends a signal to the ESP32, which then activates the corresponding vibration motors, creating a tangible sensation. This innovation enhances immersion and interactivity, making VR experiences more realistic and engaging.

2 Objectives

The primary objectives of this project are:

- Develop a pair of VR-compatible haptic gloves using mini vibration motors.
- Establish seamless wireless communication between the VR game and ESP32 module.
- Implement a real-time feedback mechanism to simulate touch sensations in VR.
- Evaluate user experience and effectiveness of haptic feedback through testing.

3 Scope of Project

The primary scope of this project is to evaluate the effect of Haptic on VR experience. I will evaluate the effect by conducting a small group study on very small group of people.
Evaluation Proposal:
After I complete the hardware part, I will create a game/course on VR using unity that can be experienced by users. My evaluation will be based on time taken by user to complete these tasks with and without Haptic gloves.

4 Game proposed

The VR game/course I am creating in Unity will be an interactive experience where users complete various tasks that require precise hand interactions. These tasks will include turning a switch on and off, placing geometric objects into their respective slots, sliding a button, and popping bubbles. To enhance engagement, I can introduce varying difficulty levels, such as objects that require different levels of force to move, time-based challenges, or tasks that require multitasking. Additionally, incorporating real-world physics, such as resistance while sliding the button or a slight delay when pressing the switch, can make interactions feel more natural. To further improve immersion, I may add visual and audio feedback, such as sound effects when objects are placed correctly.

5 Evaluation metric

To evaluate the impact of haptic gloves, I will measure the time taken by users to complete these tasks both with and without haptic feedback. The experiment will be conducted on a small group of participants, and I will average the completion times to determine if haptic feedback enhances efficiency. To strengthen the evaluation, I can also collect qualitative feedback through user surveys, asking about perceived realism, ease of interaction, and overall engagement. Additionally, tracking user behavior, such as hesitation time before interacting with objects or error rates (e.g., incorrect object placement), can provide deeper insights into the effectiveness of haptic feedback. By analyzing both quantitative and qualitative data, I will be able to assess whether haptic integration significantly enhances the user experience.

6 Methodology

The development of the vibrotactile haptic gloves will follow a structured approach, ensuring seamless integration with the VR system. The methodology consists of multiple phases:

1. Hardware Design: The gloves will be embedded with mini vibration motors at the fingertips, powered by an ESP32 microcontroller. The ESP32 will be responsible for receiving signals from the VR game and activating the appropriate motors. Power efficiency and ergonomic design will be prioritized.

2. Software Development: The VR game will be programmed to detect object interactions and send real-time feedback signals to the ESP32 via Wi-Fi or Bluetooth. A custom firmware for ESP32 will process these signals and trigger the corresponding haptic feedback on the gloves.

Figure 1: System architecture of the VR haptic gloves

3. Communication Protocol: The project will establish a stable and low-latency communication link between the VR game and the gloves. This will involve developing an optimized data transmission protocol to ensure real-time responsiveness.

4. Integration and Testing: The gloves will be tested with various VR environments to evaluate their performance in providing tactile feedback. Adjustments will be made to optimize vibration intensity, latency, and user experience.

7 Timeline

Phase	**Timeline**
Research	Completed
Hardware Completion	February 15th - March 15th
Game Creation using Unity	February 15th - March 15th
First Trial Phase	March 15th - April 1st
Update based on user feedback	April 1st - May 1st

8 Resources Required

1)Gloves
2)ESP32 WIFI Module
3)VR(META QUEST 3)
4)Mini Vibration motors